

Qinwei (Martin) Ma

First-year PhD student



personal

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IIIS, Tsinghua University

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Certificates & Grants

Gold medal in National
High School Physics
Olympiad

First Prize in Shanghai in
National High School
Informatics Olympiad

Second Prize in Shanghai
in National High School
Mathematics Olympiad

Languages

Chinese (Native)
English (Very Fluent)
German (B1 level)
Japanese (Beginner)

Interests

Musical Theater
played the role Aaron Burr
in *Hamilton*

Bridge Card
won second place in
National College Students'
Bridge Open Tournament
Mixed Division

Soccer
Core member of the
soccer team in IIIS
department (before injury).

SHORT RESUMÉ & CURRICULUM

2026–Present

College of AI, Tsinghua University
PHD STUDENT · Beijing 📍
Advised by Prof. **Alex Lamb**.



2021–2026

IIIS / Yao Class, Tsinghua University
UNDERGRADUATE STUDENT · Beijing 📍
Computer Science and Technology, with specialization in
Artificial Intelligence.



RESEARCH INTEREST

My research connects reinforcement learning for generative models with world-model-grounded decision making. I am interested in how expressive policies can learn from interaction, optimize generative objectives, and remain tied to the dynamics of the environments where they act.

- **RL for expressive generative policies:** improving flow-matching policies, generative models, and LLMs through interaction and ELBO-based objectives, with a focus on how timestep weighting shapes credit assignment, exploration, and policy improvement.
- **World-model-grounded decision making:** using learned world models to support policy learning, planning, and decision making, both by generating experience and by grounding policies through dynamics, feasibility, memory, and long-horizon consistency.

Long-term vision. I aim to build adaptive agents that improve through interaction while keeping their decisions grounded in the environments they observe, model, and influence.

ACADEMIC

BASIC INFO

Overall GPA:	3.75	TOEFL:	109
Major GPA:	3.88	GRE:	328
Citations:	132	h-index:	6
i10-index:	4		

AWARDS AND HONORS

- **Finalist, College of AI NSFC Doctoral-Student Nomination (Sep. 2026)**
One of four doctoral students selected for the college-wide final nomination defense as a first-year PhD student.
- **First Prize in Challenge Cup**
A project-based competition where I led all AI-related tasks.
- **Three School-level Scholarships**
Recognized for achievements in Scientific and Technological Innovation, Artistic and Cultural Performance and Social Work respectively.
- **Gold Medal in National High School Physics Olympiad**
Awarded in my second year of high school, directly granting me admission to Yao Class.

MAIN RESEARCH EXPERIENCE (IN REVERSE CHRONOLOGICAL ORDER)

Tencent Hunyuan (Video World Model Team) | *Student Intern* Feb. 2026 - Aug. 2026
Self-forcing, distillation, and RL post-training for video world models.

Tsinghua University (College of AI) | *PhD Student/Researcher (Advisor: Prof. Alex Lamb)* [2026 - Present]

Generative-policy RL and world-model-grounded agents.

Stanford University | *Research Intern/Ongoing Collaborator (Advisor: Prof. James Zou)* [2025 - 2026]

Physics reasoning and RL post-training; ongoing collaboration.

UIUC | *Research Intern (Advisor: Prof. Tong Zhang)* [Jan. 2025 - Jun. 2025]

Embodied VLM fine-tuning and online RL.

Other Research Experience | *Northwestern University, MIT Cocosci Lab, and University of Washington* [2023 - 2025]

Multi-agent reasoning, multimodal and embodied learning, NLP, and time-series research.

Tsinghua University (IIIS) | *Research Intern (Advisor: Prof. Hang Zhao)* [Nov. 2022 - Oct. 2023]
Deep learning and multimodal learning.

TEACHING EXPERIENCE

Tsinghua University, College of AI | *Teaching Assistant (Instructor: Prof. Alex Lamb)* Autumn 2026
Graduate course: World Models, Exploration and Actions.

Tsinghua University (IIIS) | *Teaching Assistant (Instructor: Prof. Alex Lamb)* Spring 2026
Deep Self-Supervised Learning.

Shanghai Qizhi Institute & Xiongan Institute of AI | *Teaching Assistant* Autumn 2025
Natural Language Processing; prepared materials and delivered a guest lecture in English.

SELECTED PUBLICATIONS & PREPRINTS

* stands for equal contribution, † stands for second author

- **Timestep Weighting: A Hidden Key to Effective ELBO-Based Flow-Matching RL**; *Preprint; Under review*
Qinwei Ma*, Jingzhe Shi*, Simin Fan, Ling Li, Mengdi Wang, Alex Lamb
This work studies the role of timestep weighting in ELBO-based reinforcement learning objectives for flow-matching policies.
- **Intrinsic Entropy of Context Length Scaling in LLMs**; *ICLR 2026 Oral*
Jingzhe Shi*, **Qinwei Ma***, Hongyi Liu*, Hang Zhao, Jenq-Neng Hwang, Lei Li
We propose Intrinsic Entropy as a framework for explaining how long context affects LLM performance. Theory and experiments show how dataset size governs optimal context length and bounds scaling efficiency. OpenReview / arXiv / Code
- **PRISM-Physics: Causal DAG-Based Process Evaluation for Physics Reasoning**; *ICLR 2026*
Wanjia Zhao*, **Qinwei Ma***, Jingzhe Shi*, Shirley Wu, Jiaqi Han, Yijia Xiao, Si-Yuan Chen, Xiao Luo, Ludwig Schmidt, James Zou
We introduce a process-level physics reasoning benchmark based on causal DAGs, enabling more granular evaluation than answer-only scoring. The benchmark aligns closely with expert judgments and provides a useful signal for RL finetuning. OpenReview / arXiv
- **ERA: Transforming VLMs into Embodied Agents via Embodied Prior Learning and Online Reinforcement Learning**; *Preprint*
Hanyang Chen*, Mark Zhao*, Rui Yang*, **Qinwei Ma**[†], Ke Yang, Kangrui Wang, Hao Bai, Zhenhailong Wang, Jiarui Yao, Rui Pan, Mengchao Zhang, Jose Barreiros, Aykut Onol, ChengXiang Zhai, Heng Ji, Manling Li, Huan Zhang, Tong Zhang
ERA is a two-stage framework that scales embodied vision-language agents by combining knowledge distillation with online reinforcement learning. It uses embodied priors and online RL to improve planning and manipulation in long-horizon settings.
- **Scaling Law for Time Series Forecasting**; *NeurIPS 2024*
Jingzhe Shi*, **Qinwei Ma***, Huan Ma, Lei Li
We propose a scaling-law theory for time series forecasting, explaining when larger models and longer look-back horizons fail to help. The analysis connects dataset size, model complexity, and data granularity.

ADDITIONAL PUBLICATIONS & PREPRINTS

* stands for equal contribution, † stands for second author

- **RapVerse: Coherent Vocals and Whole-Body Motions Generations from Text**; *ICCV 2025*
Jiabao Chen, Xin Yan, Yihang Chen, Siyuan Cen, Zixin Wang, **Qinwei Ma**, Haoyu Zhen, Kaizhi Qian, Lie Lu, Chuang Gan
A unified framework and dataset for jointly generating rapping vocals and 3D whole-body motion from text lyrics.
- **CHOPS: CHat with custOmer Profile Systems for Customer Service with LLMs**; *COLM 2024*
Jingzhe Shi, Jialuo Li, **Qinwei Ma**, Zaiwen Yang, Huan Ma, Lei Li
An LLM customer-service agent that integrates with existing databases and systems for accurate, safe operations.
- **Explore-Execute Chain: Towards an Efficient Structured Reasoning Paradigm**; *Preprint*
Kaisen Yang*, Lixuan He*, Rushi Shah, Kaicheng Yang, **Qinwei Ma**, Dianbo Liu, Alex Lamb. Code
A structured reasoning framework that separates exploration from execution for more efficient reasoning.
- **Gradient Imbalance in Direct Preference Optimization**; *Preprint*
Qinwei Ma, Jingzhe Shi, Can Jin, Jenq-Neng Hwang, Serge Belongie, Lei Li
This work identifies gradient imbalance in DPO and proposes Balanced-DPO as a stabilizing alternative.

ONGOING RESEARCH

- **WM-GRPO: ELBO-based Reinforcement Learning for Robotic World Models**
We are investigating how ELBO-based reinforcement learning can be applied to robotic world models, and how learned world dynamics can inform or constrain policy optimization.
- **State Space Models Provide Extra Memory for Long-Horizon World Models**
We are exploring whether state space models can supply persistent memory for long-horizon world models, especially when maintaining consistency over extended interactions.
- **PRISM-RL: RL Post-Training for Physics Reasoning**
We are exploring RL post-training recipes for difficult physics reasoning problems, with an emphasis on whether newly acquired strategies can generalize beyond the training distribution.